

Project Update Report

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Project Title: Logic Puzzle Solver

Project Group: 02

This project is being done for the Artificial Intelligence course (CSE440). The main goal of our project is to build a system that can solve logic puzzles like Sudoku automatically using AI techniques. The solver will take an incomplete Sudoku grid, follow the rules of the puzzle, and generate the correct solution. So far, my role in this project has mainly been understanding how the puzzle works, studying the AI concepts behind the solution, planning the program structure, and helping with the initial setup of the code.

Week 1 (Feb 9 – Feb 15): Understanding the Project

In the first week, I focused on understanding what our project is about and how Sudoku works as a logic puzzle. I reviewed the rules of Sudoku carefully. I learned that each number from 1 to 9 can appear only once in each row, once in each column, and once in each 3×3 box. Since these rules are strict, I understood that Sudoku can be treated as a structured problem that a computer can solve step by step.

I also tried to understand the project from an AI point of view. Instead of seeing Sudoku as just a puzzle game, I started to see it as a problem-solving task where the system has to make decisions while following certain rules. This week was important for me because it gave me a strong base for the later stages of the project.

Week 2 (Feb 16 – Feb 22): Learning the Solving Methods

During the second week, I spent time learning about the methods that can be used to solve Sudoku in AI. I studied the idea of Constraint Satisfaction Problems, or CSP. From this, I understood that each empty cell in the Sudoku board can be considered a variable, the possible numbers from 1 to 9 are the values, and the Sudoku rules act as the constraints.

I also learned about backtracking and constraint propagation. Backtracking helped me understand how the program can try a value, check whether it is valid, and go back if the choice leads to a wrong path. Constraint propagation showed me how some wrong choices can be eliminated early by using the puzzle rules. My own contribution in this week was mainly studying these ideas and understanding how they can be applied to our project.

Week 3 (Feb 23 – Mar 1): Choosing the Tools and Planning the Program

In the third week, I focused on deciding how we would implement the solver. I worked on choosing Python as the programming language because it is simple, easy to understand, and well suited for solving algorithmic and AI-related problems. It also allows faster testing during development.

After that, I planned how the Sudoku board would be stored inside the program. I decided that using a 9×9 two-dimensional array would be a practical approach. In this setup, the filled cells contain numbers and the empty cells are represented by 0. I also thought about how the solver would later check valid moves and find empty positions. This week, my contribution was mostly in planning the basic structure that would make the implementation easier.

Week 4 (Mar 2 – Mar 8): Initial Coding and Testing

In the fourth week, I started working on the early coding part of the project. I helped set up the basic program structure so that a Sudoku grid can be given as input and displayed properly. I also worked on simple functions such as finding empty cells and handling the puzzle grid correctly.

Even though this was only the starting part of implementation, it was very important because the full solver depends on these functions. I tested the early program setup with sample Sudoku boards to make sure the grid representation was working properly. My contribution during this week was mainly in building and checking these initial functions.

My Individual Contribution So Far

Since this is a group project, we have shared the overall idea and discussed the direction together. However, my own contribution so far includes understanding the Sudoku rules in detail, studying the AI techniques related to solving the puzzle, helping choose the program structure, and working on the initial code setup. I also helped with testing the early grid functions using sample inputs. These steps helped me build a clear understanding of how the complete solver should work.

Current Progress

So far, the project is still in its early stage, but the foundation has already been built. I now have a better understanding of the puzzle, the AI concepts needed for solving it, and the basic structure of the program. The initial setup of the code is ready, and the next step will be to implement the full solving algorithm and test it on different Sudoku puzzles.